Saeed Odak

Postdoctoral Researcher · Aalto University
Department of Computer Science · Espoo, Finland

saeed.odak@aalto.fi
cglab.ca/~odak

ACADEMIC APPOINTMENTS

May 2025–Present **Postdoctoral Researcher**
Algorithms and Computational Geometry Group, Aalto University · Espoo, Finland
Research on computational geometry, hyperbolic geometry, approximation schemes, fine-grained geometric lower bounds, and algorithmic graph theory.

Jan. 2021–May 2025 **Research Assistant**
Algorithms, Graphs, and Geometry Lab (AGGLAB), Carleton University · Canada
Research on algorithmic and structural graph theory, proximity data structures, graph drawing, and computational geometry.

Sept.–Nov. 2023 **Research Assistant**
LaBRI, University of Bordeaux · France
Visiting research on the Euclidean freeze-tag problem and universal induced subgraphs.

May–Aug. 2019 **Undergraduate Researcher**
K. N. Toosi University of Technology · Iran
B.Sc. thesis on Routing and Wavelength Assignment (RWA) in Optical Networks, Supervised by Lotfollah Beygi.

EDUCATION

2021–2025 **Ph.D. in Computer Science** · University of Ottawa, Canada
Supervised by P. Bose, J-L. De Carufel, V. Dujmović, P. Morin

2014–2019 **B.Sc. in Mathematics** · K. N. Toosi University of Technology, Iran

2014–2019 **B.Sc. in Electrical Engineering** · K. N. Toosi University of Technology, Iran

2010–2014 High School Diploma, Math and Physics · NODET, Gorgan, Iran

RESEARCH INTERESTS

Computational geometry Geometric streaming algorithms, Algorithms and combinatorial problems in discrete and computational geometry, including geometric spanners, proximity problems, geometric optimization, and fine-grained complexity.

Graph theory Structural and algorithmic graph theory, including product structure, graph colouring, planarity, and sparse graph classes.

Graph drawing Structural and algorithmic problems in graph drawing and beyond-planar graphs, including layered and constrained graph representations.

TEACHING & SUPERVISION

Student Supervision

2022–2024 **Marc Vicuna (M.Sc.)** · School of Computer Science, Carleton University.
Thesis: *Efficient Computation of Interesting Paths*.

Jun.–Aug. 2025 **Satu Koitto (B.Sc.)** · Department of Computer Science, Aalto University.
Thesis: *Linear and Centered Chromatic Number of Graphs*.

Jun.–Aug. 2026 **Beimar Naranjo** · Summer Intern, Aalto Science Institute (AScI), Aalto University.

Teaching

Jan.–Apr. 2023 **Contract Instructor** · Carleton University — Discrete Structures I (COMP 1805)

2021–2024 **Teaching Assistant** · University of Ottawa — Design and Analysis of Algorithms; Data Structures and Algorithms; Discrete Structures; Introduction to Computing I & II; Introduction to Formal Languages.

2016–2019 **Teaching Assistant** · K. N. Toosi University of Technology — Algorithm Design; Data Structures; Calculus I & II.

2016–2020 **Volunteer Teaching** · K. N. Toosi University of Technology — Coaching ICPC team (Algorithms and Data Structures).

HONORS & AWARDS

2023 **PhD Mobility Program in France (MDF) Research Scholarship** · University of Bordeaux

2021–2025 **International Doctorate Scholarship** · University of Ottawa

2021–2024 **Ph.D. Admission Scholarship** · University of Ottawa

2021 **29th place** · IEEE-Xtreme 15.0 International Programming Contest (team)

2019 **Ranked 2nd** among all B.Sc. students of Mathematics · K. N. Toosi University of Technology

2019 **Ranked 7th** in National Entrance Exam for M.Sc. in Computer Science · Tehran, Iran

2019 **Silver Medal** · ACM-ICPC Asia Tehran Regional Contest (team), Sharif University

2018 **Bronze Medal** · International Mathematics Competition (IMC), Blagoevgrad, Bulgaria

2018 **Silver Medal** · Iranian Mathematics Competition, Behshahr, Iran

2018 **7th place** · ACM-ICPC Asia Tehran Regional Contest (team), Sharif University

2018 **27th place** · IEEE-Xtreme 12.0 International Programming Contest (team)

2017 **Bronze Medal** · Iranian Mathematics Competition, Shahr-e Kord, Iran

INVITED TALKS

Shifting is Optimal under Gap-ETH:
A Lower Bound Framework for Geometric Approximation Schemes
@ HIIT · Aalto University, 2025.

Connected dominating sets in triangulations
@ GADLAB · University of Manitoba, 2025 & @ HIIT · Aalto University, 2025.

Tight bounds on the number of closest pairs in vertical slabs
@ AGGLAB · Carleton University, 2024.

Euclidean freeze-tag problem on the plane
@ CGLAB · Technical University of Dortmund, 2025.

An optimal algorithm for product structure in planar graphs
@ Labri · University of Bordeaux, 2023 & @ HIIT · Aalto University, 2024.

PROFESSIONAL SERVICE

Conference reviewing SoCG (2025, 2026); ESA (2026); SoSA (2026); CCCG (2022, 2023, 2024, 2025, 2026); WG (2026); GD (2024, 2026); WADS (2023, 2025); SWAT (2024, 2026); CALDAM (2024); LAGOS (2025); ISAAC (2022).

Journal reviewing *ACM Transactions on Algorithms*; *Computational Geometry: Theory and Applications*; *Discrete Applied Mathematics*; *Computing in Geometry and Topology*.

WORKSHOPS & SUMMER SCHOOLS

2026 **Organizer** of the First Aalto Computational Geometry Workshop · Aalto University, Espoo, Finland.

2025 Rhein-Ruhr Computational Geometry Workshop · FernUniversität in Hagen, Hagen, Germany.

2024 Eleventh Annual Workshop on Geometry and Graphs · Bellairs Research Institute, Holetown, Barbados

2023 Tenth Annual Workshop on Geometry and Graphs · Bellairs Research Institute, Holetown, Barbados

2023 Summer School in Convex and Discrete Geometry · Erdős Center — Alfréd Rényi Institute of Mathematics, Budapest, Hungary

2022 Summer School in Geometry and Topology in a Discrete Setting · Berlin Mathematical School, Berlin, Germany

2021 Workshop in Graph Product Structure Theory (21w5235) · Banff International Research Station (BIRS), Banff, Canada

2019 Workshop in Data Science and Combinatorial Algorithms · KNTU, Tehran, Iran

2018 Summer School in Mathematics · Institute for Advanced Studies in Basic Sciences, Zanjan, Iran

PUBLICATIONS

Refereed Conferences

[1] Prosenjit Bose, Vida Dujmović, Hussein Houdrouge, Pat Morin, and Saeed Odak. *Connected dominating sets in triangulations*. In 53rd International Colloquium on Automata, Languages, and Programming (**ICALP 2026**).

[2] Sándor Kisfaludi-Bak, Saeed Odak, Satyam Singh, and Geert van Wordragen. *Gap-ETH-Tight Algorithms for Hyperbolic TSP and Steiner Tree*. In 42nd International Symposium on Computational Geometry (**SoCG 2026**).

[3] Manuel Cáceres, Sándor Kisfaludi-Bak, and Saeed Odak. *Shifting is Optimal under Gap-ETH: A Lower Bound Framework for Geometric Approximation Schemes*. In European Symposium on Algorithms (**ESA 2026**).

[4] Sujoy Bhore, Ahmad Biniaz, Kevin Buchin, Jean-Lou De Carufel, Antonia Kalb, Anil Maheshwari, Saeed Odak, Carolin Rehs, and Michiel Smid. *Sparse Oriented Spanners in Metric Spaces*. In European Symposium on Algorithms (**ESA 2026**).

[5] Saeed Odak, Jonathan Rollin, and Torben Scheele. *On 2-Layer k-Matching-Planar Graphs*. In 34th International Symposium on Graph Drawing and Network Visualization (**GD 2026**).

[6] Ahmad Biniaz, Jean-Lou De Carufel, Anil Maheshwari, Saeed Odak, Michiel Smid. *Minimum Sum Vertex Cover via Minimum Vertex Cover*. In the 37th International Symposium on Algorithms and Computation (**ISAAC 2026**).

[7] Ahmad Biniaz, Anil Maheshwari, Magnus Christian Ring Merrild, Joseph S. B. Mitchell, Saeed Odak, Valentin Polishchuk, Eliot W. Robson, Casper Moldrup Rysgaard, Jens Kristian Refsgaard Schou, Thomas C. Shermer, Jack Spalding-Jamieson, Rolf Svenning, and Da Wei Zheng. *Polynomial-time algorithms for contiguous art gallery and related problems*. In 41st International Symposium on Computational Geometry (**SoCG 2025**).

[8] Kevin Buchin, Antonia Kalb, Anil Maheshwari, Saeed Odak, Carolin Rehs, Michiel Smid, and Sampson Wong. *Computing oriented spanners and their dilation*. In 41st International Symposium on Computational Geometry (**SoCG 2025**).

[9] Ahmad Biniaz, Prosenjit Bose, Chaeyoon Chung, Jean-Lou De Carufel, John Iacono, Anil Maheshwari, Saeed Odak, Michiel Smid, and Csaba D. Tóth. *Tight bounds on the number of closest pairs in vertical slabs*. In 19th Algorithms and Data Structures Symposium (**WADS 2025**).

[10] Greg Aloupis, Ahmad Biniaz, Prosenjit Bose, Jean-Lou De Carufel, David Eppstein, Anil Maheshwari, Saeed Odak, Michiel Smid, Csaba D. Tóth, and Pavel Valtr. *Noncrossing longest paths and cycles*. In 32nd International Symposium on Graph Drawing and Network Visualization (**GD 2024**).

[11] Michael A. Bekos, Prosenjit Bose, Aaron Büngener, Vida Dujmović, Michael Hoffmann, Michael Kaufmann, Pat Morin, Saeed Odak, and Alexandra Weinberger. *On k-planar graphs without short cycles*. In 32nd International Symposium on Graph Drawing and Network Visualization (**GD 2024**).

[12] Nicolas Bonichon, Cyril Gavoille, Nicolas Hanusse, and Saeed Odak. *Euclidean freeze-tag problem on plane*. In The 36th Canadian Conference on Computational Geometry (**CCCG 2024**).

[13] Prosenjit Bose, Pat Morin, and Saeed Odak. *An optimal algorithm for product structure in planar graphs*. In 18th Scandinavian Symposium and Workshops on Algorithm Theory (**SWAT 2022**).

Refereed Journals

[14] Ahmad Biniaz, Prosenjit Bose, Chaeyoon Chung, Jean-Lou De Carufel, John Iacono, Anil Maheshwari, Saeed Odak, Michiel Smid, and Csaba D. Tóth. *Closest pair queries in vertical slabs and tight bounds on the number of possible answers*. In the Journal of Computational Geometry (**JoCG**), 17(1):164–183, 2026.

[15] Greg Aloupis, Ahmad Biniaz, Prosenjit Bose, Jean-Lou De Carufel, David Eppstein, Anil Maheshwari, Saeed Odak, Michiel Smid, Csaba D. Tóth, and Pavel Valtr. *Noncrossing longest paths and cycles*. In the Journal of Graphs and Combinatorics, 41(6):122, 2025.

[16] Michael A. Bekos, Prosenjit Bose, Aaron Büngener, Vida Dujmović, Michael Hoffmann, Michael Kaufmann, Pat Morin, Saeed Odak, and Alexandra Weinberger. *On k-planar graphs without short cycles*. In the Journal of Graph Algorithms and Applications, 29(3):1–22, 2025.

[17] Ahmad Biniaz, Prosenjit Bose, Jean-Lou De Carufel, Anil Maheshwari, Babak Miraftab, Saeed Odak, Michiel Smid, Shakhar Smorodinsky, and Yelena Yuditsky. On separating path and tree systems in graphs. In the *Journal of Discrete Mathematics & Theoretical Computer Science*, 27(2), 2025.

Preprints

[18] Ahmad Biniaz, Anil Maheshwari, Joseph S. B. Mitchell, Saeed Odak, Valentin Polishchuk, and Thomas C. Shermer. *Contiguous boundary guarding*. `CoRR, abs/2412.15053`, 2024.

[19] Vida Dujmović, Pat Morin, and Saeed Odak. *Odd colourings of graph products*. `CoRR, abs/2202.12882`, 2022.

[20] Jean-Lou De Carufel, Anil Maheshwari, Saeed Odak, Bodhayan Roy, Michiel Smid, and Marc Vicuna. *Deciding if a DAG is interesting is hard*. Preprint. `CoRR, abs/2503.13398`, 2025.

Submitted

[21] Prosenjit Bose, Kaya Gouin, Pat Morin, Saeed Odak. *A Linear-Time Algorithm for Product Structure in Planar Graphs* . Submitted to the Journal of ACM Transactions on Algorithms (**TALG**).

[22] Sándor Kisfaludi-Bak, Saeed Odak, Satyam Singh, and Geert van Wordragen. *Gap-ETH-Tight Algorithms for Hyperbolic TSP and Steiner Tree*. Submitted to the Journal of ACM Transactions on Algorithms (**TALG**).

[23] Ahmad Biniaz, Jean-Lou De Carufel, Anil Maheshwari, Saeed Odak, André van Renssen, Michiel Smid, and Sampson Wong. *Euclidean TSP with Mandatory Returns to a River Line*. Submitted to (**MFSC**) 2026.